

2. Brute Force and Meet-in-the-Middle Attack on Double DES

Double DES is a naive attempt to improve DES security by encrypting data twice with two keys. A MitM attack can exploit this as follows:

Attack:

- Suppose $C = E_{K2}(E_{K1}(P))$, where P is the plaintext, E_{K1} is encryption with key $K1$, and E_{K2} is encryption with key $K2$.
- The attacker computes all possible values of $E_{K1}(P)$ for all $K1$ and stores them in a table.
- Simultaneously, they compute $D_{K2}(C)$ for all $K2$ and compare the results with the table.
- A match reveals both $K1$ and $K2$.

This attack reduces the effective key space of Double DES to approximately $2^{56+56} = 2^{57}$ rather than the expected 2^{112} .

Triple DES Overview

Triple DES applies the DES algorithm three times using either two or three unique keys:

- **Encrypt-Decrypt-Encrypt (EDE):** $C = E_{K3}(D_{K2}(E_{K1}(P)))$, where E is encryption and D is decryption.
- Typically, it uses:
 - **Two-key 3DES:** $K1 = K3$ (effective key length: 112 bits).
 - **Three-key 3DES:** All keys are distinct (effective key length: 168 bits).

1. MitM Attack on Double DES Doesn't Work on Triple DES

The MitM attack that works on **Double DES** (reducing its effective key space to 2^{57}) cannot be directly applied to **Triple DES** because of the additional layer of encryption/decryption. The added complexity requires significantly more computation.

For a MitM attack on 3DES, an attacker would have to perform the following:

- Compute all possible $E_{K1}(P)$ for the first encryption layer.
- Compute all possible $D_{K3}(C)$ for the third decryption layer.
- Attempt to match the results through the middle layer $D_{K2}(E_{K1}(P)) = E_{K3}(D_{K2}(C))$.

This attack would require storing a massive amount of intermediate data and performing a much larger number of comparisons, making it computationally infeasible for 2^{112} or 2^{168} key spaces.

3. Known-Plaintext Attacks

If an attacker knows a plaintext P and its corresponding ciphertext C , they might attempt a brute force MitM attack:

- Compute $E_{K1}(P)$ for all $K1$, then $D_{K3}(C)$ for all $K3$.
- Store all intermediate values and look for matches.
- The effective key space of 2^{112} for two-key 3DES or 2^{168} for three-key 3DES makes this attack impractical.